R55: complete A-zero source quotient and degree-80 CRT

The four binary degree-20 points now exhaust the A=0 scheme in the actual normal affine source. Full module products identify the quotient, all source functions, and every quotient class. The complete pole divisor remains a separate projective geometry calculation.

Full annihilator on the actual source

B is the retained normal rank-32 RR algebra over F2[z]. All 528 multiplication columns bind the unit and actual V coordinate. With $V^2+z^2V+z+z^3=0$, set

```
R=(1+z)V+z^3; A=1+z(R+1); fA=z^5+z^2+1
```

The complete 32-by-32 multiplication matrices satisfy

```
C=M_A+(z^3+z^4)I; M_A C=C M_A=fA I32
```

Thus fA B is contained in A B. Reduction modulo fA adds no extra constraint to B/A B. A changed conjugate matrix is detected. Source: quotient55-02.

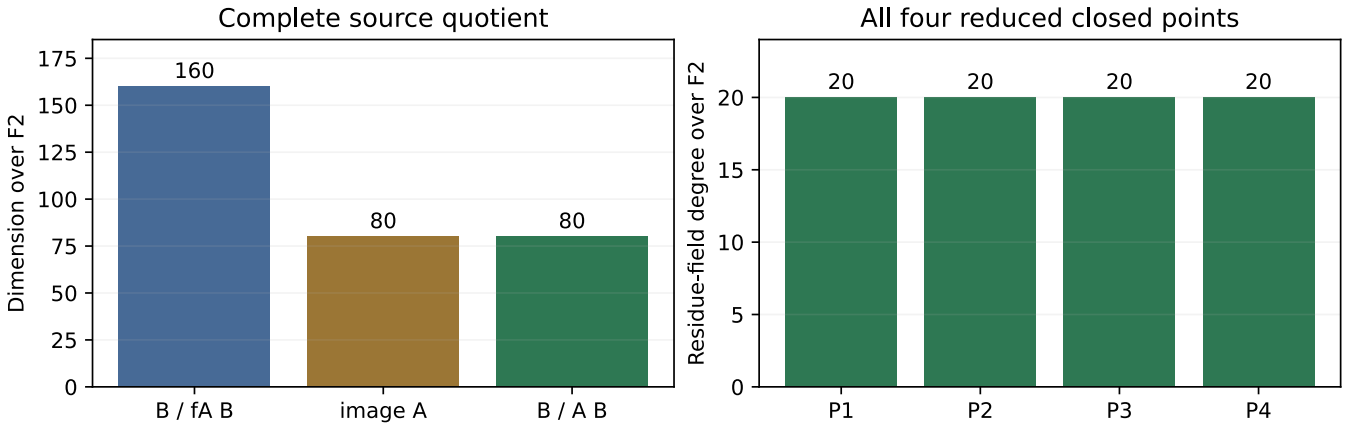
Actual normal source	Complete binary source	Exact quotient recovery	Binary CRT algebra
32 generators over F2[z]	160 coordinates modulo fA	E G = I80	Degree-80 squarefree modulus
All 528 multiplication columns	Abar has rank 80	Abar H = I160 - G E	Four degree-20 fields
A A-prime = fA I	E Abar = 0	No omitted quotient classes	All 32 source values

Exact kernel and recovery products

B/fA B has 160 binary coordinates. Abar has rank 80 and E has rank 80. The retained arrays give all three exact products

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E Abar=0; E G=I80; Abar H=I160-G E
```

The last product proves ker(E)=image(Abar) for every source class, while the second proves surjectivity. Consequently B/A B is exactly the proposed 80-dimensional algebra. This is a full source quotient identification, not just the rank of a sample or a determinant prediction.



All plotted dimensions are exact native data, without sampling uncertainty. Source: quotient55-02. The point degrees are the four retained R54 fields.

Complete binary functions, proofs and retained failures

Four fields and every source function

The product N of the four degree-20 moduli has degree 80 and is squarefree. Four full cofactor/Bezout products construct CRT idempotents; all 16 orthogonality products, all 16 residue labels and partition of unity pass. CRT recovers z=rho(w) and all 32 normal RR basis functions. The recovered base coordinate satisfies fA(rho)=0. The actual unit and all 528 source multiplication identities pass modulo N.

Thus B/A B is the product of four degree-20 fields. Its four closed points have local length one and total binary length 80 in this normal affine chart. The fresh serialized checker reconstructs every binary and evaluation entry, all recovery products, all 128 point-value labels and all 528 source products. An altered recovery matrix is rejected. Source: check55-02.

Lean claim fidelity

Four authored implications compile and pass the separate axiom audit: surjectivity from E G=I, exact kernel/image recovery, submodule equality, and scalar multiples lying in image A from A C=f I. Zero failing and zero trusted declarations. Actual matrix arrays, polynomial arithmetic, irreducible fields and retained source normalization remain external premises. See PROOF_MAP.md.

Complete workflow ledger

Workflow	Status	Receipt
check55-01	rejected	exit 0; 12 error lines
check55-02	accepted	exit 0; 0 error lines
diag55-01	accepted	exit 0; 0 error lines
quotient-lean55-01	accepted	compile 0; audit 0; four declarations
quotient55-01	rejected	exit 0; 359 error lines
quotient55-02	accepted	exit 0; 0 error lines

The first quotient attempt is rejected: a combined native scalar/matrix expression produced a 32-by-64 matrix and failed the base identity. Later undefined objects and a late marker are excluded. A separately frozen shape diagnostic reproduced the issue. The first checker attempt also retained a ring-local variable declaration failure; its output is rejected. Entrywise scalar construction passes the original formulas and checks. No failed output supports the quotient.

Native limits: 4 GiB virtual, 1,200 seconds, one Singular/FLINT thread, seed 55. Lean 4.34.1 / mathlib d13f23b, -j1/-M4096, separate 1,200-second compile/audit caps, no virtual cap. Historical executed bytes are checked against Git and drivers execute immutable copies. No accepted producer is rerun for custody. CPU times were not collected; no comparative performance claim is made.

Next geometry and preserved lanes

Determine all infinity branches, residue fields, ramification and A pole orders, then bind this affine zero scheme to the complete projective divisor. Use the degree-80 evaluator for an entire finite modular-map control before generic target identities. Preserve the actual degree-8/128 level-11 map and intended-11 action, full incidence saturation, proper polarization, node187, arbitrary reverse classes and the larger degree-112/991 source lane.

Sources: R50/R51 full RR algebra, R52 PID embedding, R53 base geometry, R54 four point fields, Stacks00DT CRT, native linalg.lib. See CM.md, protocol.json, results.json and PROOF_MAP.md for exact premises and receipts.